Tim Beauchamp

Engineering Leadership and Applied AI

Belmont, California | tim@beauchampfamily.org | 650 333 1267
linkedin.com/in/tbeauchamp | github.com/timbeauchamp

Professional Profile

Engineering manager, developer, and inventor with 35 years in software, including more than 22 years at Adobe. Leads teams building AI and traditional software for Mac, Windows, mobile devices, and the web. Uses AI in software design, development, testing, deployment, and day-to-day management. Background includes communications, distributed systems, statistical software, and databases.

Interested in building interoperable, distributed data communication systems, with applications in environmental monitoring, nowcasting, and remote telemetry.

Adobe

Engineering Manager / Computer Scientist | 2004-Present | San Jose, California

Work includes a mobile and desktop video editor, an iOS and Android editing SDK, and a web-native video editor. Current responsibilities include AI product development and the use of AI throughout the engineering process.

AI Engineering and Product Direction

- Manage developers building AI and traditional software for native and web platforms, using AI throughout design, development, testing, and deployment.
- Help engineers adopt development workflows in which specifications direct AI agents. Set expectations for requirements, code review, testing, ownership, and the tools the team needs.
- Started a recurring engineering forum to share lessons, problems, and techniques as the team moved to a new platform and agentic development.
- Contribute to design decisions for generated media, including where models run, how previews work, and how assets reach the final output. Proposed reusable editing timelines as operations in a graph-based workflow.
- Work on connecting AI tools and generative AI artifacts to traditional software through MCP, including requirements for telemetry and error reporting.

People Leadership and Engineering Management

- Coach engineers, give direct feedback, support promotions, and explain performance criteria so people understand what is expected and how to grow.
- Help teams adapt to changing products and responsibilities. Recognized by colleagues for building a welcoming, respectful, collaborative team.
- Use AI to summarize communications, extract meeting actions, check sprint consistency, and prepare regression retrospectives.

Adobe Delivery and Technical Work

Content Credentials and Release Coordination

- Coordinated Content Credentials rollout to all users of a web-native video editor in August 2026, bringing together client, media services, export, QA, program management, legal, and communications teams.
- Received management recognition for bringing the Content Credentials work together and leading delivery across teams.
- Resolve release dependencies with product, design, engineering, QA, and service teams, including ownership, testing, and support after launch.

Performance Analysis and Engineering Tools

- Investigated media ingest performance with Splunk, Python, and AI-assisted analysis, separating media processing costs from concurrency and queue delays.
- Traced upload delays to cached-block registration overhead and HTTP throttling. Proposed changes to request pacing, batching, and service limits.
- Designed a measurement framework covering upload preparation, transfer, processing, readiness, download, and display, accounting for cache state and differences between workloads.
- Built a Python/FFmpeg test-media generator and a blind audio-comparison service to evaluate time-stretch quality.
- Identified duplicate work and release risk caused by differences between playback and export. Called for shared architectural planning and earlier involvement from service teams.

Cross Platform Products and Release Engineering

- Managed engineering for a professional video editor on mobile and desktop platforms, with releases through app stores and cloud services.
- Led a team developing an iOS and Android video-editing SDK; helped prepare a web-native video editor for public beta.
- Diagnosed native build and release problems involving Jenkins, Python packaging, and iOS frameworks. Added switching between test and production environments to the iOS video editor.
- Organized innovation weeks with project proposals, volunteer teams, check-ins, and demos.

Tools and Technologies

AI and integration: Claude, Cursor, ChatGPT, MCP, and agentic development. Analysis and systems: Python, SQL, Splunk, HTTP, APIs, media pipelines, benchmarking, and telemetry. Additional experience: Java, Kotlin, Swift, JavaScript, C/C++, Objective-C, native mobile applications, databases, and data warehousing.

Independent Projects and Earlier Experience

Selected Independent Projects

Project Phaethon - Built a sensing prototype linking camera nodes and irradiance measurements to cloud services, archived observations, and dashboards. Developed Python/FastAPI services and alert evaluation against sensor data. Nowcasting performance is under evaluation.

WhatPlaneIsThat - Built Android and iOS aircraft-identification software using ADS-B data, device location and orientation, geographic calculations, and camera overlays. Related work includes OpenDroneID parsing and orientation and coordinate-conversion libraries.

Network Wires - Built sensor firmware, AWS Lambda/DynamoDB services, and web and Android interfaces. Includes software and hardware prototyping, IoT, and telemetry.

Local AI and automation - Built a Python gateway that routes tasks and reports timing, errors, uncertainty, and which capabilities are available. Integrated home automation software, MQTT, voice services, Linux-based embedded computing hosts, InfluxDB, and Grafana.

Patents

Named co-inventor on 12 granted U.S. software patents covering real-time enterprise intelligence, event-driven collaboration, resilient networked applications, and adaptive user interfaces. Work originated at Iteration/Istante Software; grants were assigned to Oracle. Selected examples:

US 7,941,542 B2 - Maintaining application execution over an intermittent network connection.
US 8,402,095 B2 - Instant messaging collaboration.
US 8,001,185 B2 - Distributed rule evaluation in near real-time business intelligence.

Earlier Career

Istante Software - Founder / Senior Software Developer, real-time business intelligence; acquired by Oracle. AOL-Netscape - Senior Software Engineer, browser development. Data Digest - first engineer, statistical analysis software. Sagent - data warehousing development and QA. NetManage - TCP/IP and Internet application development and QA. Software Publishing Corporation - Harvard Graphics.

Additional experience includes developing Bayesian statistical analysis and modeling software, document classification systems, databases and data warehouses, and 3D modeling and rendering.

Education and Additional Experience

Electronic Engineering - California Polytechnic State University, San Luis Obispo.
General Maintenance Mechanic - Antarctic Support Associates, Amundsen-Scott South Pole Station, Antarctica.